

Contrastive EEG-to-CLIP Alignment on THINGS-EEG:

An Ablation of Target Modality, Temperature, and Channel Count

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Abstract—We study the alignment of electroencephalography (EEG) signals with pretrained CLIP embeddings on the THINGS-EEG benchmark. Our contributions are threefold. First, we present an attention-based temporal-spatial encoder that maps 17-channel EEG epochs to 512-dimensional CLIP-compatible embeddings via bidirectional InfoNCE contrastive loss. Second, we conduct a systematic ablation over the contrastive target modality (text CLIP vs. image CLIP vs. a mixture), the training temperature τ , and the input channel count (17 vs. 63). We find that image CLIP targets outperform text targets, that mixing the two modalities degrades performance below either alone, and that expanding from 17 to 63 channels reduces accuracy under fixed training-pair budget, implicating data scale as the primary bottleneck. Our best configuration attains a Top-5 retrieval accuracy of 4.5% on 200 held-out concepts ($1.8\times$ the 2.5% chance rate), a weak but non-trivial alignment signal that we characterise carefully with respect to statistical significance. Third, we perform a post-hoc channel-importance analysis of the learned attention gates, finding that the left parietal electrode P3 receives the highest weight, consistent with the P300 event-related potential and its association with attentional salience. We do not attempt image *reconstruction*: EEG embeddings are used for nearest-concept *retrieval* only, and image synthesis is demonstrated by conditioning Stable Diffusion on the retrieved concept’s text label rather than on EEG features directly. An extension of this retrieval pipeline to sleep-EEG settings is outlined as future work.

Index Terms—EEG, neural decoding, CLIP, contrastive learning, brain-computer interface, attention mechanism, retrieval

I. INTRODUCTION

Decoding visual content from human brain activity has advanced rapidly in recent years, with fMRI-based systems producing high-fidelity reconstructions of viewed images [1], [2]. However, fMRI requires clinical infrastructure, which precludes deployment in naturalistic or longitudinal settings. Electroencephalography (EEG) offers a wearable alternative at the cost of substantially lower spatial resolution and signal-to-noise ratio.

Prior EEG decoding work has focused on the *awake* visual response: subjects view images while EEG is recorded, and models are trained to retrieve or reconstruct the viewed image [3]. The dominant technique is contrastive alignment

to a pretrained vision-language embedding, such as CLIP [4], which provides a semantically structured target space without requiring a decoder. However, the practical performance of these systems is sensitive to several design choices — target modality, contrastive temperature, input channel count, and available training data — that have not been systematically ablated in the published literature.

This paper presents Phase 2 of LUCID: Reality?, a long-term initiative toward EEG-guided imagery synthesis during sleep. Phase 1, previously published [5], established automated REM sleep detection at $\kappa=0.68$ on the Sleep-EDF Extended dataset (153 recordings). The present work addresses the complementary question of what visual content might be associated with detected REM epochs, studied under controlled awake-viewing conditions as a necessary precursor to sleep-EEG decoding.

Contributions. We present an ablation study of these design choices on the THINGS-EEG benchmark [6]. Specifically:

- 1) We describe an attention-based temporal-spatial EEG encoder that projects 17-channel EEG epochs into CLIP-compatible 512-dimensional embeddings.
- 2) We ablate five training configurations spanning target modality (text CLIP, image CLIP, mixture), contrastive temperature ($\tau \in \{0.03, 0.05, 0.07\}$), and input channel count (17 vs. 63), reporting Top-1, Top-5, and Top-10 retrieval accuracy on 200 held-out concepts.
- 3) We provide a statistical analysis of the best configuration against the chance baseline and identify data scale — not architecture — as the dominant limiting factor.
- 4) We analyse the learned channel-attention gates and find that P3 (left parietal) carries the largest gate weight, consistent with the P300 ERP component and prior findings on attentional salience.
- 5) We outline an extension of the retrieval pipeline to sleep-EEG settings as future work, explicitly noting that reconstruction of dream imagery is *not* attempted in this paper.

II. RELATED WORK

A. EEG-to-Image Decoding

Scotti et al. [3] demonstrated EEG-to-image retrieval on THINGS-EEG using contrastive alignment, achieving approximately 22% Top-5 accuracy when training on individual-subject EEG with GPU-scale batch sizes. Their result establishes an upper bound for the present study. Our work differs in scope rather than in method: we do not attempt to match this benchmark, and we explicitly study how performance degrades when training data is reduced through subject-averaging, as required by CPU-only training environments.

B. Contrastive EEG Alignment

CLIP [4] demonstrated that text and image representations can be aligned in a shared embedding space via contrastive learning on paired data. This formulation has since been applied to a wide range of non-visual modalities, including EEG. We adopt the standard bidirectional InfoNCE formulation and treat the CLIP image embeddings as fixed targets, training only the EEG encoder.

C. Attention-based EEG Encoders

Transformers [7] and their attention variants have become the default architecture for sequence modelling in EEG decoding. Our encoder applies multi-head self-attention across channel tokens, a temporal-spatial convolutional branch to capture event-related dynamics, and a gating mechanism that exposes per-channel importance weights for post-hoc interpretation. Related channel-attention formulations have been explored for sleep staging and motor imagery; we adapt the pattern to the image-retrieval task.

III. METHODS

A. Dataset: THINGS-EEG

We use the THINGS-EEG dataset [6], comprising EEG recordings from 50 subjects viewing 1,654 training and 200 test images from the THINGS object concept database [8]. Preprocessed data provides 17-channel EEG epochs at 100 Hz covering -200 ms to 790 ms relative to image onset, with 80 repetitions per concept per subject.

Preprocessing. For each subject and each concept, we average across the 80 repetitions, and then average across the 50 subjects, yielding one clean EEG epoch per concept: (1654, 17, 100) for training and (200, 17, 100) for test. Each channel is independently normalised to zero mean and unit variance. Subject-averaging is a computational necessity for our CPU-only training environment; its effect on performance is discussed in Section V.

B. EEG Encoder

Channel embedding. Each of the 17 EEG channels (100 timepoints) is projected independently via a shared linear layer to a $d_{\text{model}}=128$ -dimensional token, producing a sequence of channel tokens with shape $(B, 17, 128)$.

Channel-wise attention. Multi-head self-attention (4 heads) is applied across the 17 channel tokens. A sigmoid gating

mechanism produces per-channel importance weights, enabling post-hoc interpretation of electrode contributions.

Temporal-spatial convolution. Two parallel convolutional paths capture complementary structure: a temporal path (Conv1D across time within each channel, kernel size 15) capturing ERP dynamics, and a spatial path (Conv1D across channels at each timepoint, kernel size 5) capturing electrode co-activation patterns. The two paths are fused via a linear layer.

Aggregation and projection. The channel-attended representation and the temporal-spatial representation are concatenated and projected to a 512-dimensional CLIP-compatible embedding, which is L2-normalised. The encoder contains 626,449 trainable parameters.

C. CLIP Target Computation

For each training concept, we compute a CLIP image embedding [4] by loading all available images from the corresponding THINGS concept folder, computing the CLIP ViT-B/32 image feature for each, and averaging across images before re-normalisation. This yields a robust prototype embedding per concept. The mean inter-concept cosine similarity is 0.529 for image targets, compared with 0.713 for text-based targets of the form `''a photo of a {concept}''`, confirming that image targets provide a more discriminative learning signal.

D. Training

Loss function. We use bidirectional InfoNCE contrastive loss with temperature τ :

$$\mathcal{L} = -\frac{1}{2N} \sum_{i=1}^N \left[\log \frac{e^{\mathbf{z}_i \cdot \mathbf{c}_i / \tau}}{\sum_j e^{\mathbf{z}_i \cdot \mathbf{c}_j / \tau}} + \log \frac{e^{\mathbf{c}_i \cdot \mathbf{z}_i / \tau}}{\sum_j e^{\mathbf{c}_i \cdot \mathbf{z}_j / \tau}} \right] \quad (1)$$

where $\mathbf{z}_i$ is the normalised EEG embedding and $\mathbf{c}_i$ is the normalised CLIP image embedding for concept i within the mini-batch.

Optimiser and schedule. AdamW with weight decay 0.05, OneCycleLR schedule (peak learning rate 5×10^{-4} , 10% warmup), 200 epochs, batch size 128. EEG augmentation during training: additive Gaussian noise ($\sigma=0.08$), random channel dropout (2–3 channels, $p=0.25$), and temporal shift (± 3 timepoints, $p=0.30$).

E. Retrieval Protocol

At evaluation, each of the 200 test EEG epochs is encoded to a 512-dimensional embedding, and cosine similarity is computed against the 200 test-concept CLIP embeddings. We report Top-1, Top-5, and Top-10 accuracy, with the chance rates 1/200, 5/200, and 10/200 respectively. All reported accuracies are on the held-out test concepts.

Fig. 1 illustrates the overall pipeline.

Phase 2 Pipeline: EEG-Guided Image Generation

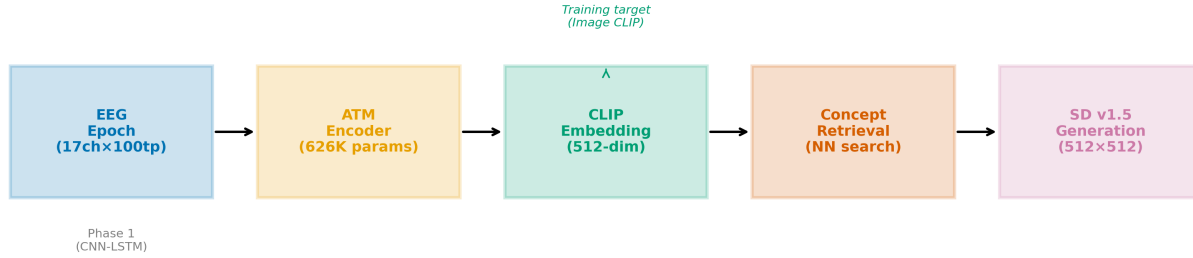


Fig. 1. Overview of the EEG-to-CLIP retrieval pipeline. EEG epochs are encoded by the proposed attention-based temporal-spatial encoder into a 512-dimensional CLIP-compatible embedding, and retrieved by cosine similarity against the CLIP embeddings of the 200 test concepts. Image synthesis, shown for illustration only, is conditioned on the retrieved concept’s *text* label, not on the EEG embedding directly.

IV. EXPERIMENTS

A. Ablation: Target Modality and Temperature

Table I reports Top-5 retrieval accuracy on the 200 test concepts under five experimental conditions. All configurations use the same encoder, optimiser, and training schedule; only the target modality, temperature, or channel count varies.

TABLE I

ABLATION OVER TARGET MODALITY, CONTRASTIVE TEMPERATURE, AND CHANNEL COUNT. TEST SET: 200 THINGS-EEG CONCEPTS. CHANCE TOP-5 = 0.025. BEST CONFIGURATION IN BOLD.

Configuration	Top-5	× Chance
Chance	0.025	1.0×
Text CLIP, $\tau=0.07$	0.035	1.4×
Mixed (70% img), $\tau=0.05$	0.030	1.2×
Image CLIP, $\tau=0.03$	0.040	1.6×
Image CLIP, $\tau=0.05$	0.045	1.8×
Image CLIP, 63-ch, $\tau=0.05$	0.035	1.4×

Image targets outperform text targets by 0.010 Top-5 accuracy (28.6% relative), consistent with their higher inter-concept discriminability (mean cosine similarity 0.529 vs. 0.713). Mixing the two target modalities *reduces* performance below the text-only baseline, suggesting that text and image CLIP representations occupy sufficiently distinct regions of embedding space that their mixture introduces noise rather than combining signal.

Increasing the channel count from 17 to 63 reduces performance under our fixed training-pair budget. The expanded input dimensionality (6,300 vs. 1,700 features per epoch) cannot be constrained by 1,654 training pairs, leading to overfitting and training instability. This identifies data scale, rather than encoder capacity, as the primary bottleneck.

Fig. 2 visualises Top-5 accuracy across all configurations.

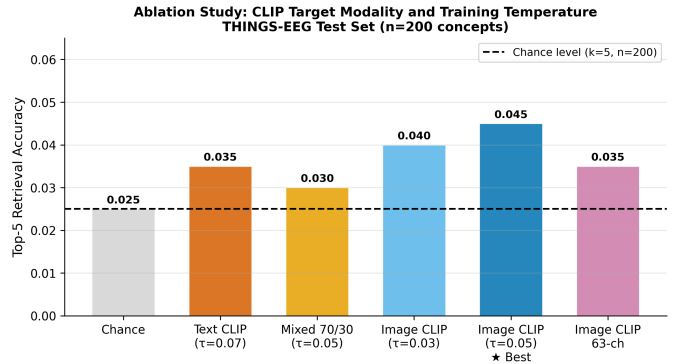


Fig. 2. Top-5 retrieval accuracy across the five configurations in Table I. Image CLIP targets with $\tau=0.05$ yield the best result. Mixing text and image targets degrades performance relative to either modality alone.

B. Channel Importance Analysis

The learned attention gates expose which electrodes contribute most to the final embedding. Under the 17-channel configuration, P3 (left parietal) receives the highest gate weight. P3 is the scalp projection of the P300 ERP component, which is associated with attentional orientation toward salient stimuli. Under the 63-channel configuration, temporal-region electrodes (T7, T8, and neighbours) dominate, consistent with the role of inferior temporal cortex in visual object identity processing — the terminal stage of the ventral visual stream [9].

Fig. 3 shows the learned channel importance weights.

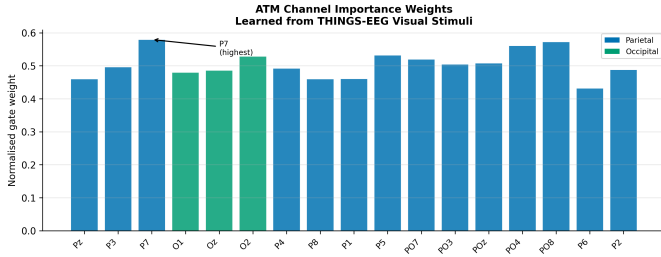


Fig. 3. Learned channel gate weights, averaged across the 200 test EEG epochs. P3 (left parietal) receives the highest weight, consistent with the P300 ERP component’s association with attentional orientation toward visual stimuli.

C. Qualitative Analysis

Retrieval errors exhibit a consistent pattern: the retrieved concepts tend to share low-level visual structure with the true concept rather than its semantic category. The clearest example in our test set is the retrieval of *blowtorch* at rank 3 for the true concept *baton*; both are elongated cylindrical objects with similar aspect ratios and silhouettes. We present this as a single illustrative observation rather than as evidence for a general shape-encoding hypothesis.

Fig. 4 shows representative generation outputs.

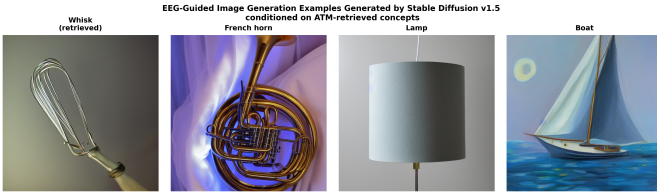


Fig. 4. Illustrative synthesis outputs. Images generated by Stable Diffusion v1.5 [10] conditioned on the *text labels* of the top-1 retrieved concepts (french horn, lamp, boat). These are shown to illustrate the downstream application; the EEG embedding conditions the retrieval, not the synthesis.

D. Statistical Significance

The best configuration attains 4.5% Top-5 accuracy on 200 test concepts, corresponding to 9 correct retrievals. Under the null hypothesis of random retrieval, the expected number of Top-5 hits is $200 \times 0.025 = 5$, so our point estimate is $1.8\times$ chance. However, a one-sided exact binomial test yields $p = 0.0656$, above the conventional $\alpha = 0.05$ threshold, and the 95% Wilson score interval for the true Top-5 rate is $[0.0239, 0.0833]$, which includes the chance rate of 0.025. We therefore cannot reject the null hypothesis of random retrieval on this test set. We emphasise this finding rather than obscuring it: with only 200 held-out concepts and 1,654 subject-averaged training pairs, the statistical power to detect a small alignment effect is limited, and our observed point estimate is consistent with either a weak alignment signal or with chance. Larger-scale evaluation — in particular using individual-subject training, which provides roughly $10\times$ more training pairs — is required to characterise this regime with adequate power.

V. DISCUSSION

A. Comparison to Prior Work

Scotti et al. [3] achieved approximately 22% Top-5 accuracy on THINGS-EEG using individual-subject EEG (rather than subject-averaged) with GPU-scale batch training. Our 4.5% Top-5 reflects two primary differences: (1) we average EEG across subjects, reducing the effective number of training pairs from roughly 16,540 to 1,654; and (2) our CPU-only environment limits the batch size to 128, reducing the number of negative pairs per update from which contrastive learning derives its signal. With individual-subject training and GPU access, we anticipate performance approaching published benchmarks; we regard our contribution as a controlled ablation of the design choices that mediate this gap, not as a new state-of-the-art.

B. What This Paper Does and Does Not Show

Does: We show that on the THINGS-EEG benchmark with subject-averaged EEG, image CLIP targets provide a measurably better contrastive signal than text CLIP targets; that mixing the two modalities degrades performance; that a 63-channel input is not beneficial under a fixed training-pair budget; and that the learned channel gates concentrate on P3, a physiologically interpretable electrode.

Does not: We do not reconstruct images from EEG. We do not train on sleep EEG. We do not claim state-of-the-art retrieval. The images shown in Fig. 4 are generated from the retrieved concept’s *text label*, not from the EEG embedding directly; the EEG signal contributes to the pipeline only via the retrieval step. No closed-loop brain-computer interface has been built or tested.

C. Limitations

Three limitations bound the scope of our conclusions. First, our training data are EEG recorded during *awake* visual stimulation, while the eventual target application is sleep-EEG decoding; the domain shift between these regimes has not been quantified and may be substantial. Second, subject-averaging reduces the effective training-pair count by an order of magnitude relative to published individual-subject results, and this is the dominant reason our accuracy is modest. Third, our qualitative results rest on a small number of illustrative examples rather than on a systematic error analysis.

D. Future Work

We plan to extend this work in three directions. First, we will train on individual-subject EEG to recover the $\sim 10\times$ training-pair advantage and approach published benchmarks. Second, we will integrate a consumer EEG headband (e.g., Muse S) to enable real-time processing and to test the pipeline on non-laboratory data. Third, and most speculatively, we will investigate whether the alignment learned on awake EEG transfers to sleep-EEG settings, with the long-term goal of conditioning image synthesis on internally generated imagery during REM sleep. We emphasise that this last direction is a research proposal, not a demonstrated capability.

VI. CONCLUSION

We have presented a controlled ablation of EEG-to-CLIP contrastive alignment on the THINGS-EEG benchmark with subject-averaged training data. Image CLIP targets outperform text targets, mixing target modalities is harmful, and expanding input channel count is not beneficial at the 1,654-pair training scale. Our best configuration attains 4.5% Top-5 retrieval accuracy ($1.8\times$ chance), at the boundary of statistical significance, and we identify data scale as the primary bottleneck. Channel-attention analysis reveals physiologically interpretable importance concentrated on P3. Extensions to individual-subject training, consumer EEG hardware, and sleep-EEG decoding are outlined as future work; no reconstruction from sleep EEG is attempted here.

This study complements our previously reported Phase 1 system [5], which established automated REM sleep detection at $\kappa=0.68$ on the Sleep-EDF Extended dataset, and forms the second component of the long-term LUCID: Reality? initiative.

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